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## Local feature: Corners

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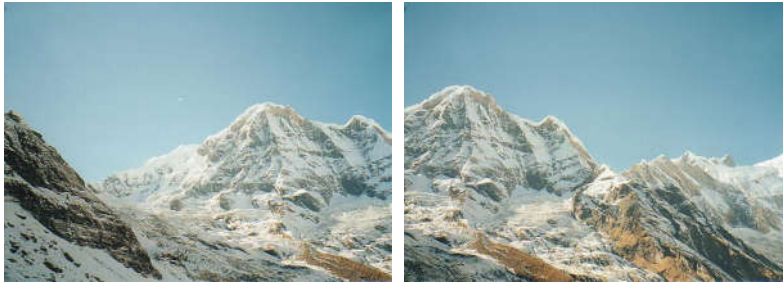
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| Machine Vision Technology                                    |                            |                          |                    |                       |                       |                      |     |
|--------------------------------------------------------------|----------------------------|--------------------------|--------------------|-----------------------|-----------------------|----------------------|-----|
| Semantic information                                         |                            |                          |                    | Metric 3D information |                       |                      |     |
| Pixels                                                       | Segments                   | Images                   | Videos             | Camera                |                       | Multi-view Geometry  |     |
| Convolutions<br>Edges & Fitting<br>Local features<br>Texture | Segmentation<br>Clustering | Recognition<br>Detection | Motion<br>Tracking | Camera<br>Model       | Camera<br>Calibration | Epipolar<br>Geometry | SFM |
| 10                                                           | 4                          | 4                        | 2                  | 2                     | 2                     | 2                    | 2   |

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## Why extract features?

- Motivation: panorama stitching
  - We have two images – how do we combine them?



Source: S. Lazebnik

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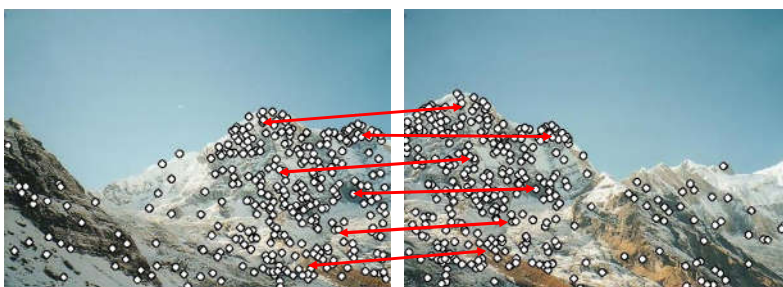
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## Why extract features?

- Motivation: panorama stitching
  - We have two images – how do we combine them?



Step 1: extract features

Step 2: match features

Source: S. Lazebnik

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## Why extract features?

- Motivation: panorama stitching
  - We have two images – how do we combine them?



Step 1: extract features

Step 2: match features

Step 3: align images

Source: S. Lazebnik

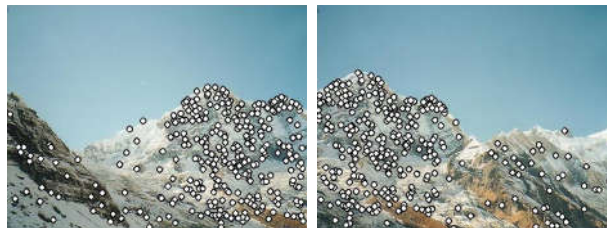
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## Characteristics of good features



- Repeatability
  - The same feature can be found in several images despite geometric and photometric transformations
- Saliency
  - Each feature is distinctive
- Compactness and efficiency
  - Many fewer features than image pixels
- Locality
  - A feature occupies a relatively small area of the image; robust to clutter and occlusion

Source: S. Lazebnik

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## Applications

Feature points are used for:

- Image alignment
- 3D reconstruction
- Motion tracking
- Robot navigation
- Indexing and database retrieval
- Object recognition



Source: S. Lazebnik

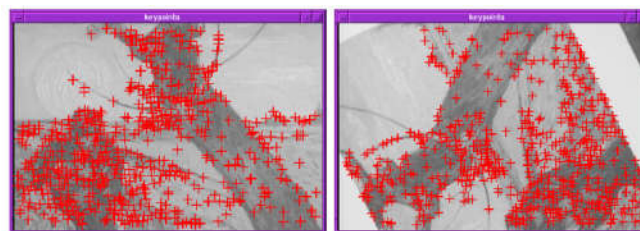
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## Finding Corners



- Key property: in the region around a corner, image gradient has two or more dominant directions
- Corners are repeatable and distinctive

C.Harris and M.Stephens. ["A Combined Corner and Edge Detector."](#)  
*Proceedings of the 4th Alvey Vision Conference*: pages 147--151.

Source: S. Lazebnik

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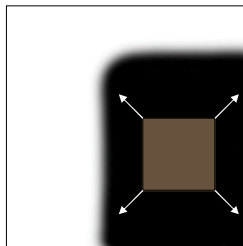
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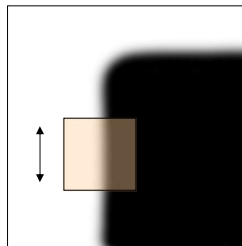
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## Corner Detection: Basic Idea

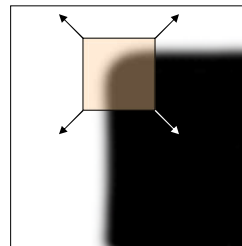
- We should easily recognize the point by looking through a small window
- Shifting a window in *any direction* should give a *large change* in intensity



"flat" region:  
no change in all  
directions



"edge":  
no change along the  
edge direction



"corner":  
significant change  
in all directions

Source: A. Efros

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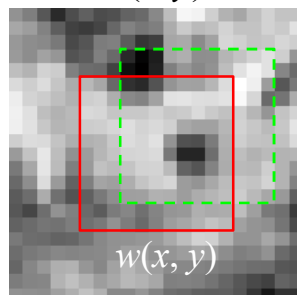
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## Corner Detection: Mathematics

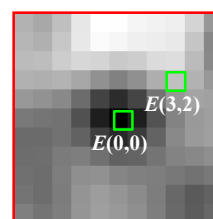
Change in appearance of window  $w(x,y)$   
for the shift  $[u,v]$ :

$$E(u,v) = \sum_{x,y} w(x,y) [I(x+u, y+v) - I(x,y)]^2$$

$I(x,y)$



$E(u,v)$



Source: S. Lazebnik

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## Corner Detection: Mathematics

Change in appearance of window  $w(x,y)$   
for the shift  $[u,v]$ :

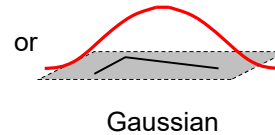
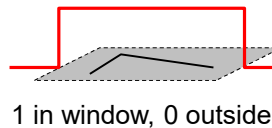
$$E(u,v) = \sum_{x,y} w(x,y) [I(x+u, y+v) - I(x,y)]^2$$

Window  
function

Shifted  
intensity

Intensity

Window function  $w(x,y) =$



Source: R. Szeliski

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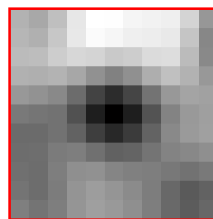
## Corner Detection: Mathematics

Change in appearance of window  $w(x,y)$   
for the shift  $[u,v]$ :

$$E(u,v) = \sum_{x,y} w(x,y) [I(x+u, y+v) - I(x,y)]^2$$

We want to find out how this function behaves for  
small shifts

$E(u, v)$



Source: S. Lazebnik

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## Corner Detection: Mathematics

Change in appearance of window  $w(x,y)$   
for the shift  $[u,v]$ :

$$E(u,v) = \sum_{x,y} w(x,y) [I(x+u, y+v) - I(x,y)]^2$$

We want to find out how this function behaves for small shifts

Local quadratic approximation of  $E(u,v)$  in the neighborhood of  $(0,0)$  is given by the *second-order Taylor expansion*:

$$E(u,v) \approx E(0,0) + [u \ v] \begin{bmatrix} E_u(0,0) \\ E_v(0,0) \end{bmatrix} + \frac{1}{2} [u \ v] \begin{bmatrix} E_{uu}(0,0) & E_{uv}(0,0) \\ E_{uv}(0,0) & E_{vv}(0,0) \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

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## Corner Detection: Mathematics

$$E(u,v) = \sum_{x,y} w(x,y) [I(x+u, y+v) - I(x,y)]^2$$

Second-order Taylor expansion of  $E(u,v)$  about  $(0,0)$ :

$$E(u,v) \approx E(0,0) + [u \ v] \begin{bmatrix} E_u(0,0) \\ E_v(0,0) \end{bmatrix} + \frac{1}{2} [u \ v] \begin{bmatrix} E_{uu}(0,0) & E_{uv}(0,0) \\ E_{uv}(0,0) & E_{vv}(0,0) \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

$$E_u(u,v) = \sum_{x,y} 2w(x,y) [I(x+u, y+v) - I(x,y)] I_x(x+u, y+v)$$

$$E_{uu}(u,v) = \sum_{x,y} 2w(x,y) I_x(x+u, y+v) I_x(x+u, y+v) + \sum_{x,y} 2w(x,y) [I(x+u, y+v) - I(x,y)] I_{xx}(x+u, y+v)$$

$$E_{uv}(u,v) = \sum_{x,y} 2w(x,y) I_y(x+u, y+v) I_x(x+u, y+v) + \sum_{x,y} 2w(x,y) [I(x+u, y+v) - I(x,y)] I_{xy}(x+u, y+v)$$

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## Corner Detection: Mathematics

$$E(u, v) = \sum_{x, y} w(x, y) [I(x+u, y+v) - I(x, y)]^2$$

Second-order Taylor expansion of  $E(u, v)$  about  $(0, 0)$ :

$$E(u, v) \approx E(0, 0) + [u \ v] \begin{bmatrix} E_u(0, 0) \\ E_v(0, 0) \end{bmatrix} + \frac{1}{2} [u \ v] \begin{bmatrix} E_{uu}(0, 0) & E_{uv}(0, 0) \\ E_{uv}(0, 0) & E_{vv}(0, 0) \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

$$\begin{array}{ll} E(0, 0) = 0 & E_{uu}(0, 0) = \sum_{x, y} 2w(x, y) I_x(x, y) I_x(x, y) \\ E_u(0, 0) = 0 & E_{vv}(0, 0) = \sum_{x, y} 2w(x, y) I_y(x, y) I_y(x, y) \\ E_v(0, 0) = 0 & E_{uv}(0, 0) = \sum_{x, y} 2w(x, y) I_x(x, y) I_y(x, y) \end{array}$$

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## Corner Detection: Mathematics

The quadratic approximation simplifies to

$$E(u, v) \approx [u \ v] M \begin{bmatrix} u \\ v \end{bmatrix}$$

where  $M$  is a *second moment matrix* computed from image derivatives:

$$M = \sum_{x, y} w(x, y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

$$M = \begin{bmatrix} \sum I_x I_x & \sum I_x I_y \\ \sum I_x I_y & \sum I_y I_y \end{bmatrix} = \sum \begin{bmatrix} I_x \\ I_y \end{bmatrix} [I_x \ I_y] = \sum \nabla I (\nabla I)^T$$

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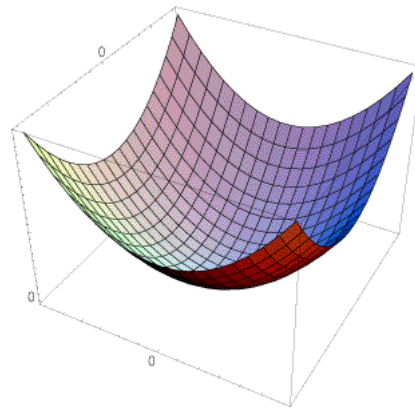


## Interpreting the second moment matrix

The surface  $E(u,v)$  is locally approximated by a quadratic form. Let's try to understand its shape.

$$E(u,v) \approx [u \ v] M \begin{bmatrix} u \\ v \end{bmatrix}$$

$$M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$



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## Interpreting the second moment matrix

First, consider the axis-aligned case (gradients are either horizontal or vertical)

$$M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

If either  $\lambda$  is close to 0, then this is **not** a corner, so look for locations where both are large.

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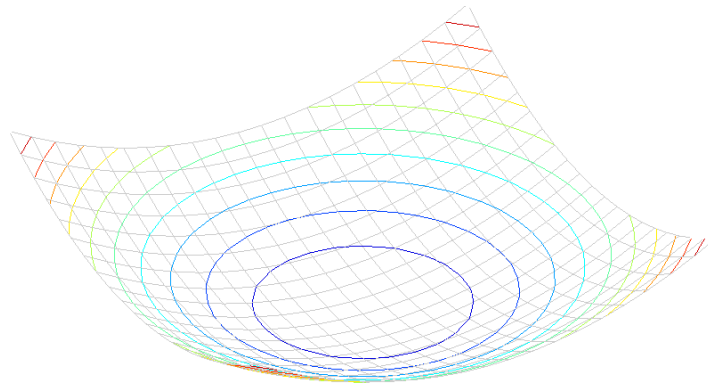
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## Interpreting the second moment matrix

Consider a horizontal “slice” of  $E(u, v)$ :  $[u \ v] M \begin{bmatrix} u \\ v \end{bmatrix} = \text{const}$

This is the equation of an ellipse.



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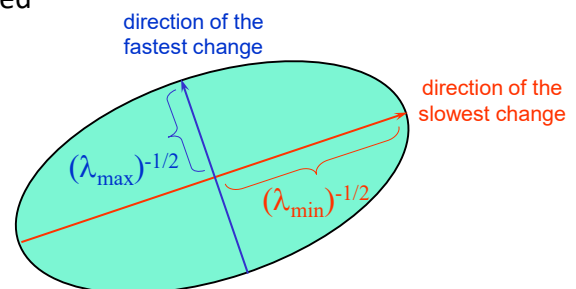
## Interpreting the second moment matrix

Consider a horizontal “slice” of  $E(u, v)$ :  $[u \ v] M \begin{bmatrix} u \\ v \end{bmatrix} = \text{const}$

This is the equation of an ellipse.

Diagonalization of  $M$ :  $M = R^{-1} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} R$

The axis lengths of the ellipse are determined by the eigenvalues and the orientation is determined by  $R$



Source: S. Lazebnik

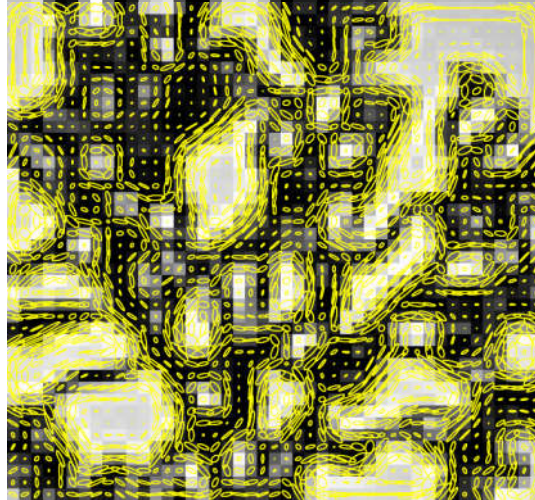
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## Visualization of second moment matrices



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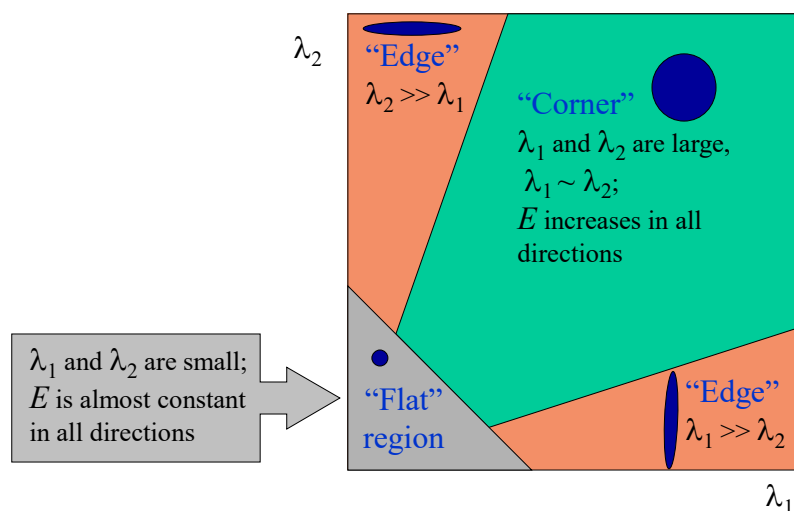
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## Interpreting the eigenvalues

Classification of image points using eigenvalues of  $M$ :



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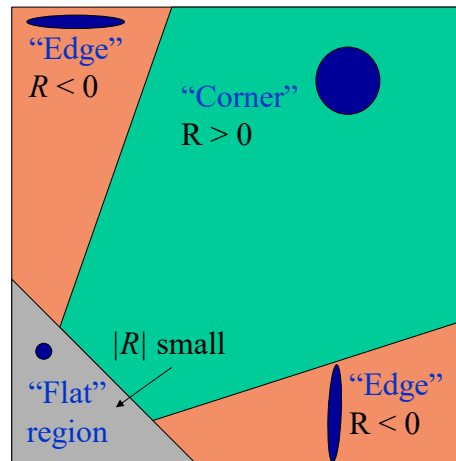
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## Corner response function

$$R = \det(M) - \alpha \text{trace}(M)^2 = \lambda_1 \lambda_2 - \alpha(\lambda_1 + \lambda_2)^2$$

$\alpha$ : constant (0.04 to 0.06)



Source: S. Lazebnik

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## Harris detector: Steps

1. Compute Gaussian derivatives at each pixel
2. Compute second moment matrix  $M$  in a Gaussian window around each pixel
3. Compute corner response function  $R$
4. Threshold  $R$
5. Find local maxima of response function (nonmaximum suppression)

C.Harris and M.Stephens. ["A Combined Corner and Edge Detector."](#)  
*Proceedings of the 4th Alvey Vision Conference*: pages 147—151, 1988.

Source: S. Lazebnik

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## Harris Detector: Steps

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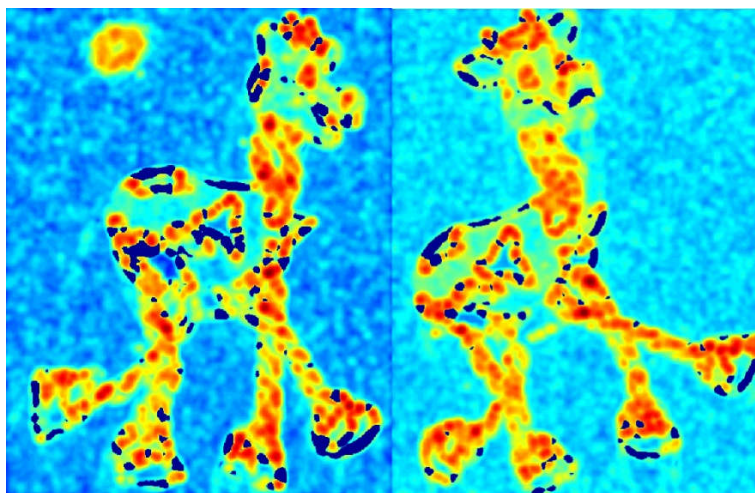
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## Harris Detector: Steps

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Compute corner response  $R$



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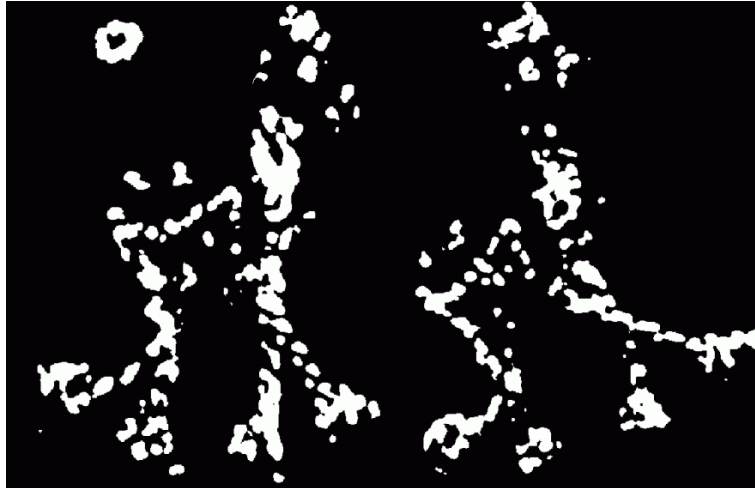
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## Harris Detector: Steps

Find points with large corner response:  $R > \text{threshold}$



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## Harris Detector: Steps

Take only the points of local maxima of  $R$



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## Harris Detector: Steps



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## Invariance and covariance

- We want corner locations to be *invariant* to photometric transformations and *covariant* to geometric transformations
  - **Invariance:** image is transformed and corner locations do not change
  - **Covariance:** if we have two transformed versions of the same image, features should be detected in corresponding locations



Source: S. Lazebnik

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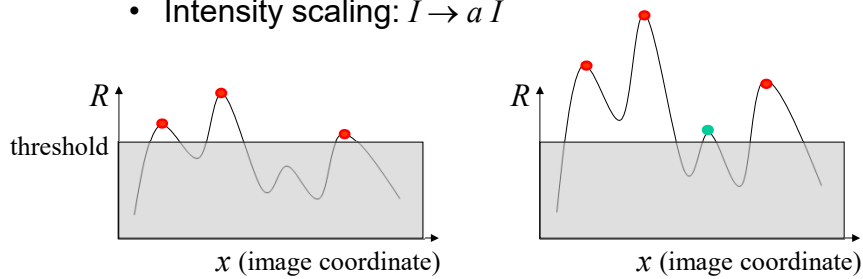
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## Affine intensity change



$$I \rightarrow aI + b$$

- Only derivatives are used => invariance to intensity shift  $I \rightarrow I + b$
- Intensity scaling:  $I \rightarrow aI$



*Partially invariant to affine intensity change*

Source: S. Lazebnik

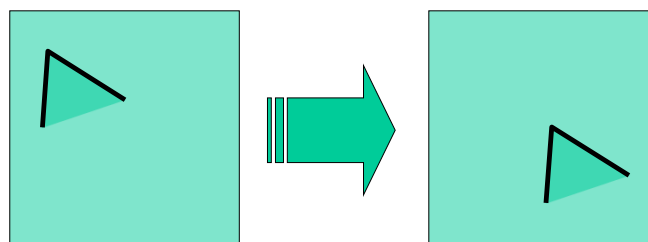
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## Image translation



- Derivatives and window function are shift-invariant

*Corner location is covariant w.r.t. translation*

Source: S. Lazebnik

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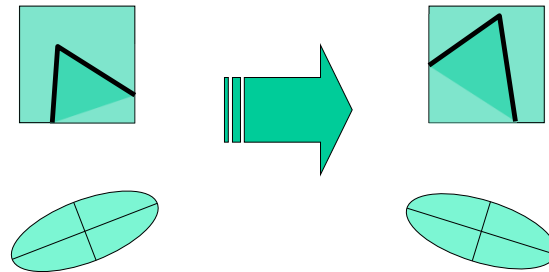
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## Image rotation



Second moment ellipse rotates but its shape  
(i.e. eigenvalues) remains the same

Corner location is covariant w.r.t. rotation

Source: S. Lazebnik

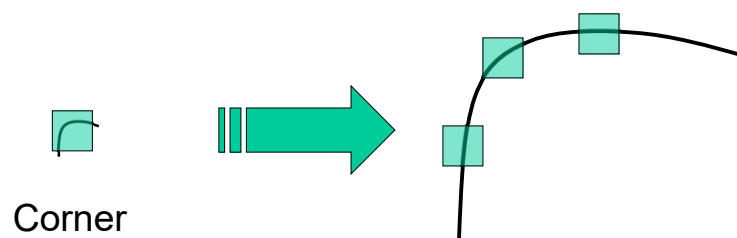
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## Scaling



All points will  
be classified  
as **edges**

Corner location is not covariant to scaling!

Source: S. Lazebnik

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